

Artificial Intelligence - Sample PDF

This sample PDF is intended for practicing Retrieval-Augmented Generation (RAG) with LangChain.

What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computer science focused on creating systems that can perform tasks that normally require human intelligence, such as learning, reasoning, problem solving, perception, and language understanding.

Machine Learning

Machine Learning is a subset of AI in which models learn patterns from data instead of being explicitly programmed.

Deep Learning

Deep Learning uses neural networks with multiple layers to solve complex problems such as image recognition and natural language processing.

Natural Language Processing (NLP)

NLP enables computers to understand, interpret, and generate human language. Large Language Models (LLMs) are a modern application of NLP.

Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with a language model. First, relevant document chunks are retrieved from a vector database. Then, the language model uses those chunks as context to generate an accurate answer.

Vector Database

A vector database stores embeddings, which are numerical representations of text. Similar vectors are located using similarity search.